

Manufactured Dynamic Solution for a Compressible Neo-Hookean Solid

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1 Dynamic problem in the reference configuration

The solid is formulated in the reference configuration. Material coordinates are written as $\mathbf{X}$ and the current position is:

$$\mathbf{x}(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t), \quad (1)$$

where $\mathbf{u}$ is the displacement field.

The balance of linear momentum in material form is:

$$\rho_0 \ddot{\mathbf{u}} = \nabla_0 \cdot \mathbf{P} + \mathbf{f}_0, \quad (2)$$

where ρ_0 is the reference mass density, $\mathbf{P}$ is the first Piola–Kirchhoff stress, and $\mathbf{f}_0$ is the body force per unit reference volume. Therefore, for a prescribed exact displacement field,

$$\mathbf{f}_0(\mathbf{X}, t) = \rho_0 \ddot{\mathbf{u}}^{\text{ex}} - \nabla_0 \cdot \mathbf{P}(\mathbf{u}^{\text{ex}}). \quad (3)$$

All external boundaries are Dirichlet boundaries,

$$\mathbf{u} = \mathbf{u}^{\text{ex}} \quad \text{on } \partial\Omega. \quad (4)$$

No traction boundary condition is required for this manufactured problem.

2 Neo-Hookean model with isochoric–volumetric split

2.1 Finite-deformation kinematics

The deformation gradient $\mathbf{F}$, Jacobian J , right Cauchy–Green tensor $\mathbf{C}$, left Cauchy–Green tensor $\mathbf{B}$, and Green–Lagrange strain $\mathbf{E}$ are

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \mathbf{I} + \nabla_0 \mathbf{u}, \quad J = \det(\mathbf{F}) > 0, \quad (5)$$

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{B} = \mathbf{F} \mathbf{F}^T, \quad (6)$$

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}). \quad (7)$$

2.2 Strain-energy density and derivation of the stress measures

The strain-energy density is additively split into isochoric and volumetric parts,

$$\Psi = \Psi_{\text{iso}} + \Psi_{\text{vol}}, \quad (8)$$

with

$$\Psi_{\text{iso}} = \frac{\mu}{2}(\bar{I}_1 - 3), \quad \Psi_{\text{vol}} = \frac{\kappa}{4}(J^2 - 1 - 2 \ln J), \quad (9)$$

where¹

$$\bar{I}_1 = J^{-2/3} I_1 \quad \text{and} \quad I_1 = \text{tr } \mathbf{C} = \text{tr } \mathbf{B}. \quad (10)$$

Thus

$$\Psi(\mathbf{F}) = \frac{\mu}{2} \left(J^{-2/3} I_1 - 3 \right) + \frac{\kappa}{4} (J^2 - 1 - 2 \ln J). \quad (11)$$

Derivation of the stress measures Since

$$\det(\mathbf{C}) = \det(\mathbf{F}^T \mathbf{F}) = \det(\mathbf{F}^T) \det(\mathbf{F}) = J^2 \underbrace{\Rightarrow}_{J>0} \boxed{J = \sqrt{\det(\mathbf{C})}}, \quad (12)$$

$$\boxed{\frac{\partial(J^{-2/3})}{\partial J} = -\frac{2}{3} J^{-5/3}}, \quad (13)$$

$$J = (\det(\mathbf{C}))^{1/2} \Rightarrow \frac{\partial J}{\partial \mathbf{C}} = \frac{1}{2} (\det(\mathbf{C}))^{-1/2} \frac{\partial \det(\mathbf{C})}{\partial \mathbf{C}} = \frac{1}{2} (\det(\mathbf{C}))^{1/2} \underbrace{\mathbf{C}^{-T}}_{\mathbf{C}=\mathbf{C}^T \Rightarrow \mathbf{C}^{-1}=\mathbf{C}^{-T}} \Rightarrow \boxed{\frac{\partial J}{\partial \mathbf{C}} = \frac{J}{2} \mathbf{C}^{-1}}, \quad (14)$$

$$\underbrace{\frac{\partial \det(\mathbf{C})}{\partial \mathbf{C}} = \det(\mathbf{C}) \mathbf{C}^{-T}}_{\text{Classical identity.}}$$

$$I_1 = \text{tr}(\mathbf{C}) = C_{11} + C_{22} + C_{33} \Rightarrow \frac{\partial I_1}{\partial C_{ij}} = \frac{\partial C_{kk}}{\partial C_{ij}} = \delta_{ij} \Rightarrow \boxed{\frac{\partial I_1}{\partial \mathbf{C}} = \mathbf{I}}. \quad (15)$$

Then the second Piola–Kirchhoff stress $\mathbf{S}$ is:

$$\begin{aligned} \mathbf{S} = 2 \frac{\partial \Psi}{\partial \mathbf{C}} &= 2 \frac{\partial \left(\frac{\mu}{2} (J^{-2/3} I_1 - 3) + \frac{\kappa}{4} (J^2 - 1 - 2 \ln J) \right)}{\partial \mathbf{C}}, \\ &= \mu \frac{\partial (J^{-2/3} I_1)}{\partial \mathbf{C}} + \frac{\kappa}{2} \frac{\partial (J^2 - 2 \ln J)}{\partial \mathbf{C}}, \\ &= \mu \left(\frac{\partial (J^{-2/3})}{\partial J} \frac{\partial J}{\partial \mathbf{C}} I_1 + J^{-2/3} \frac{\partial I_1}{\partial \mathbf{C}} \right) + \frac{\kappa}{2} \frac{\partial (J^2 - 2 \ln J)}{\partial J} \frac{\partial J}{\partial \mathbf{C}}, \\ &= \mu \left(\left(-\frac{2}{3} J^{-5/3} \right) \left(\frac{J}{2} \mathbf{C}^{-1} \right) I_1 + J^{-2/3} \mathbf{I} \right) + \frac{\kappa}{2} \left(2J - \frac{2}{J} \right) \frac{J}{2} \mathbf{C}^{-1}, \end{aligned} \quad (16)$$

therefore

$$\boxed{\mathbf{S} = \mu J^{-2/3} \left(\mathbf{I} - \frac{I_1}{3} \mathbf{C}^{-1} \right) + \frac{\kappa}{2} (J^2 - 1) \mathbf{C}^{-1} = \mu J^{-2/3} \mathbf{I} + \left(\frac{\kappa}{2} (J^2 - 1) - \mu J^{-2/3} \frac{I_1}{3} \right) \mathbf{C}^{-1}}. \quad (17)$$

The first Piola–Kirchhoff stress is

$$\boxed{\mathbf{P} = \mathbf{F} \mathbf{S} = \mu J^{-2/3} \left(\mathbf{F} - \frac{I_1}{3} \mathbf{F}^{-T} \right) + \frac{\kappa}{2} (J^2 - 1) \mathbf{F}^{-T} = \mu J^{-2/3} \mathbf{F} + \left(\frac{\kappa}{2} (J^2 - 1) - \mu J^{-2/3} \frac{I_1}{3} \right) \mathbf{F}^{-T}}. \quad (18)$$

The Kirchhoff stress is

$$\boxed{\boldsymbol{\tau} = \mathbf{P} \mathbf{F}^T = \mu J^{-2/3} \left(\mathbf{B} - \frac{I_1}{3} \mathbf{I} \right) + \frac{\kappa}{2} (J^2 - 1) \mathbf{I} = \mu J^{-2/3} \mathbf{B} + \left(\frac{\kappa}{2} (J^2 - 1) - \mu J^{-2/3} \frac{I_1}{3} \right) \mathbf{I}}. \quad (19)$$

¹In terms of Young's modulus E and Poisson's ratio ν ,

$$\mu = \frac{E}{2(1+\nu)}, \quad \kappa = \frac{E}{3(1-2\nu)} = \kappa = \lambda + \frac{2}{3}\mu \leftarrow \text{Lamé parameters.}$$

Finally,

$$\boxed{\boldsymbol{\sigma} = \frac{1}{J} \boldsymbol{\tau} = \mu J^{-\frac{5}{3}} \left(\mathbf{B} - \frac{I_1}{3} \mathbf{I} \right) + \frac{\kappa}{2} \left(\frac{J^2 - 1}{J} \right) \mathbf{I} = \mu J^{-\frac{5}{3}} \mathbf{B} + \left(\frac{\kappa}{2} \left(\frac{J^2 - 1}{J} \right) - \mu J^{-\frac{5}{3}} \frac{I_1}{3} \right) \mathbf{I}.} \quad (20)$$

Other expressions for $\boldsymbol{\tau}$ and $\boldsymbol{\sigma}$ are

$$\boldsymbol{\tau} = \mu \operatorname{dev}(\overline{\mathbf{B}}) + \frac{\kappa}{2} (J^2 - 1) \mathbf{I} \quad \text{and} \quad \boldsymbol{\sigma} = \frac{\mu}{J} \operatorname{dev}(\overline{\mathbf{B}}) + \frac{\kappa}{2} \left(\frac{J^2 - 1}{J} \right) \mathbf{I}, \quad (21)$$

where

$$\operatorname{dev}(\overline{\mathbf{B}}) = \overline{\mathbf{B}} - \frac{1}{3} \operatorname{tr}(\overline{\mathbf{B}}) \mathbf{I} \quad \text{and} \quad \overline{\mathbf{B}} = J^{-2/3} \mathbf{B}. \quad (22)$$

Defining

$$\boxed{m = \mu J^{-2/3} \quad \text{and} \quad c = \frac{\kappa}{2} (J^2 - 1) - m \frac{I_1}{3}}, \quad (23)$$

then the compact form of stresses is:

$$\boxed{\mathbf{S} = m \mathbf{I} + c \mathbf{C}^{-1}, \quad \mathbf{P} = m \mathbf{F} + c \mathbf{F}^{-\mathrm{T}}, \quad \boldsymbol{\tau} = m \mathbf{B} + c \mathbf{I} \quad \text{and} \quad \boldsymbol{\sigma} = \frac{m}{J} \mathbf{B} + \frac{c}{J} \mathbf{I}.} \quad (24)$$

3 Manufactured displacement

Suppose that the manufactured field has the form

$$\mathbf{u}(\mathbf{X}, t) = \mathbf{a} q(\mathbf{X}, t), \quad (25)$$

where $\mathbf{a} = [a_x, a_y, a_z]^{\mathrm{T}}$ is constant, therefore $u_i = a_i q$. Define

$$\mathbf{g} = \nabla_0 q, \quad A = \mathbf{a} \cdot \mathbf{a}, \quad G = \mathbf{g} \cdot \mathbf{g}. \quad (26)$$

Then

$$\nabla_0 \mathbf{u} = \nabla_0 (\mathbf{a} q) \Rightarrow \frac{\partial u_i}{\partial X_j} = a_i \frac{\partial q}{\partial X_j} = a_i g_j \Rightarrow \boxed{\nabla_0 \mathbf{u} = \mathbf{a} \mathbf{g}^{\mathrm{T}} = \mathbf{a} \otimes \mathbf{g}}, \quad (27)$$

therefore

$$\boxed{\mathbf{F} = \mathbf{I} + \mathbf{a} \mathbf{g}^{\mathrm{T}} = \mathbf{I} + \mathbf{a} \otimes \mathbf{g}}. \quad (28)$$

For the determinant we can use the identity $\det(\mathbf{A} + \mathbf{u} \mathbf{v}^{\mathrm{T}}) = \det(\mathbf{A})(1 + \mathbf{v}^{\mathrm{T}} \mathbf{A}^{-1} \mathbf{u})$, therefore:

$$J = \det(\mathbf{F}) = \det(\mathbf{I} + \mathbf{a} \mathbf{g}^{\mathrm{T}}) = \det(\mathbf{I})(1 + \mathbf{g}^{\mathrm{T}} \mathbf{I} \mathbf{a}) = 1 + \mathbf{g}^{\mathrm{T}} \mathbf{a} \Rightarrow \boxed{J = 1 + \mathbf{g}^{\mathrm{T}} \mathbf{a} = 1 + \mathbf{a} \cdot \mathbf{g}}. \quad (29)$$

The Sherman–Morrison identity is

$$(\mathbf{A} + \mathbf{u} \mathbf{v}^{\mathrm{T}})^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1} \mathbf{u} \mathbf{v}^{\mathrm{T}} \mathbf{A}^{-1}}{1 + \mathbf{v}^{\mathrm{T}} \mathbf{A}^{-1} \mathbf{u}}, \quad (30)$$

therefore:

$$\mathbf{F}^{-1} = (\mathbf{I} + \mathbf{a} \mathbf{g}^{\mathrm{T}})^{-1} = \mathbf{I}^{-1} - \frac{\mathbf{I}^{-1} \mathbf{a} \mathbf{g}^{\mathrm{T}} \mathbf{I}^{-1}}{1 + \mathbf{g}^{\mathrm{T}} \mathbf{I}^{-1} \mathbf{a}} = \mathbf{I} - \frac{\mathbf{a} \mathbf{g}^{\mathrm{T}}}{1 + \mathbf{g}^{\mathrm{T}} \mathbf{a}} = \mathbf{I} - \frac{\mathbf{a} \mathbf{g}^{\mathrm{T}}}{J} = \mathbf{I} - \frac{\mathbf{a} \otimes \mathbf{g}}{J}, \quad (31)$$

then

$$\boxed{\mathbf{F}^{-1} = \mathbf{I} - \frac{\mathbf{a} \mathbf{g}^{\mathrm{T}}}{J} = \mathbf{I} - \frac{\mathbf{a} \otimes \mathbf{g}}{J}, \quad \mathbf{F}^{-\mathrm{T}} = \mathbf{I} - \frac{\mathbf{g} \mathbf{a}^{\mathrm{T}}}{J} = \mathbf{I} - \frac{\mathbf{g} \otimes \mathbf{a}}{J}}. \quad (32)$$

The strain tensors are

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = (\mathbf{I} + \mathbf{g} \mathbf{a}^T) (\mathbf{I} + \mathbf{a} \mathbf{g}^T) = \mathbf{I} + \mathbf{g} \mathbf{a}^T + \mathbf{a} \mathbf{g}^T + \mathbf{g} \mathbf{a}^T \mathbf{a} \mathbf{g}^T = \mathbf{I} + \mathbf{g} \mathbf{a}^T + \mathbf{a} \mathbf{g}^T + \mathbf{g} (\mathbf{a} \cdot \mathbf{a}) \mathbf{g}^T \quad (33)$$

$$\mathbf{B} = \mathbf{F} \mathbf{F}^T = (\mathbf{I} + \mathbf{a} \mathbf{g}^T) (\mathbf{I} + \mathbf{g} \mathbf{a}^T) = \mathbf{I} + \mathbf{a} \mathbf{g}^T + \mathbf{g} \mathbf{a}^T + \mathbf{a} \mathbf{g}^T \mathbf{g} \mathbf{a}^T = \mathbf{I} + \mathbf{a} \mathbf{g}^T + \mathbf{g} \mathbf{a}^T + \mathbf{a} (\mathbf{g} \cdot \mathbf{g}) \mathbf{a}^T \quad (34)$$

therefore

$$\boxed{\mathbf{C} = \mathbf{I} + \mathbf{g} \mathbf{a}^T + \mathbf{a} \mathbf{g}^T + A \mathbf{g} \mathbf{g}^T = \mathbf{I} + \mathbf{g} \otimes \mathbf{a} + \mathbf{a} \otimes \mathbf{g} + A \mathbf{g} \otimes \mathbf{g}}, \quad (35)$$

$$\boxed{\mathbf{B} = \mathbf{I} + \mathbf{a} \mathbf{g}^T + \mathbf{g} \mathbf{a}^T + G \mathbf{a} \mathbf{a}^T = \mathbf{I} + \mathbf{a} \otimes \mathbf{g} + \mathbf{g} \otimes \mathbf{a} + G \mathbf{a} \otimes \mathbf{a}}, \quad (36)$$

and

$$\begin{aligned} \mathbf{C}^{-1} &= (\mathbf{F}^T \mathbf{F})^{-1} = \mathbf{F}^{-1} \mathbf{F}^{-T} = \left(\mathbf{I} - \frac{\mathbf{a} \mathbf{g}^T}{J} \right) \left(\mathbf{I} - \frac{\mathbf{g} \mathbf{a}^T}{J} \right) = \mathbf{I} - \frac{\mathbf{a} \mathbf{g}^T}{J} - \frac{\mathbf{g} \mathbf{a}^T}{J} + \frac{\mathbf{a} \mathbf{g}^T \mathbf{g} \mathbf{a}^T}{J} \\ &= \mathbf{I} - \frac{\mathbf{a} \mathbf{g}^T}{J} - \frac{\mathbf{g} \mathbf{a}^T}{J} + \frac{1}{J^2} \mathbf{a} (\mathbf{g} \cdot \mathbf{g}) \mathbf{a}^T = \mathbf{I} - \left(\frac{\mathbf{a} \mathbf{g}^T + \mathbf{g} \mathbf{a}^T}{J} \right) + \frac{(\mathbf{g} \cdot \mathbf{g})}{J^2} \mathbf{a} \mathbf{a}^T, \end{aligned} \quad (37)$$

therefore

$$\boxed{\mathbf{C}^{-1} = \mathbf{I} - \left(\frac{\mathbf{a} \mathbf{g}^T + \mathbf{g} \mathbf{a}^T}{J} \right) + \frac{G}{J^2} \mathbf{a} \mathbf{a}^T = \mathbf{I} - \left(\frac{\mathbf{a} \otimes \mathbf{g} + \mathbf{g} \otimes \mathbf{a}}{J} \right) + \frac{G}{J^2} \mathbf{a} \otimes \mathbf{a}}. \quad (38)$$

In three dimensions, including the plane-strain embedding used later,

$$\boxed{I_1 = \text{tr}(\mathbf{C}) = 3 + 2 \mathbf{a} \cdot \mathbf{g} + A G}. \quad (39)$$

Define also

$$\boxed{d = m + c \quad \text{and} \quad e = \frac{c}{J}}, \quad (40)$$

and using Eqs. (24), (28) and (32), the first Piola stress becomes

$$\boxed{\mathbf{P} = d \mathbf{I} + m \mathbf{a} \mathbf{g}^T - e \mathbf{g} \mathbf{a}^T = d \mathbf{I} + m \mathbf{a} \otimes \mathbf{g} - e \mathbf{g} \otimes \mathbf{a}}. \quad (41)$$

4 Exact material divergence and manufactured body force

For later use define

$$\boxed{\mathbf{H} = \nabla_0 \mathbf{g}, \quad \mathbf{h} = \mathbf{H} \mathbf{a} = \nabla_0 J, \quad \mathbf{r} = \mathbf{H} \mathbf{g}}, \quad (42)$$

and the following equalities

$$\boxed{\nabla \cdot (\mathbf{x} \mathbf{y}^T) = (\nabla \mathbf{x}) \mathbf{y} + \mathbf{x} (\nabla \cdot \mathbf{y})}, \quad (43)$$

$$\boxed{\nabla (\mathbf{x} \cdot \mathbf{y}) = (\nabla \mathbf{x})^T \mathbf{y} + (\nabla \mathbf{y})^T \mathbf{x}}. \quad (44)$$

$$\boxed{\nabla (\alpha \mathbf{x}) = \mathbf{x} (\nabla \alpha)^T + \alpha \nabla \mathbf{x}}. \quad (45)$$

Then the divergence of $\mathbf{P}$ is:

$$\begin{aligned} \nabla_0 \cdot \mathbf{P} &= \nabla_0 \cdot (d \mathbf{I} + m \mathbf{a} \mathbf{g}^T - e \mathbf{g} \mathbf{a}^T), \\ &= \nabla_0 \cdot (d \mathbf{I}) + \nabla_0 \cdot (m \mathbf{a} \mathbf{g}^T) - \nabla_0 \cdot (e \mathbf{g} \mathbf{a}^T), \\ &= \nabla_0 d + \nabla_0 \cdot (m \mathbf{a} \mathbf{g}^T) - \nabla_0 \cdot (e \mathbf{g} \mathbf{a}^T), \\ &= \nabla_0 d + \nabla_0 (m \mathbf{a}) \mathbf{g} + m \mathbf{a} \nabla_0 \cdot \mathbf{g} - \nabla_0 (e \mathbf{g}) \mathbf{a} - e \mathbf{g} \underbrace{\nabla_0 \cdot \mathbf{a}}_{=0}, \\ &= \nabla_0 d + \underbrace{\mathbf{a} (\nabla_0 m)^T \mathbf{g}}_{\mathbf{g} \cdot \nabla_0 m} + \underbrace{m (\nabla_0 \mathbf{a}) \mathbf{g}}_{=0} + m \mathbf{a} \underbrace{\nabla_0 \cdot \mathbf{g}}_{=\Delta_0 q} - \mathbf{g} \underbrace{(\nabla_0 e)^T \mathbf{a}}_{\mathbf{a} \cdot \nabla_0 e} - e \underbrace{(\nabla_0 \mathbf{g}) \mathbf{a}}_{=\mathbf{h}}, \end{aligned} \quad (46)$$

Therefore:

$$\boxed{\nabla_0 \cdot \mathbf{P} = \nabla_0 d + \mathbf{a}(\mathbf{g} \cdot \nabla_0 m + m \Delta_0 q) - \mathbf{g}(\mathbf{a} \cdot \nabla_0 e) - e \mathbf{h}}. \quad (47)$$

Consequently,

$$\boxed{\mathbf{f}_0 = \rho_0 \mathbf{a} \ddot{q} - \nabla_0 d - \mathbf{a}(\mathbf{g} \cdot \nabla_0 m + m \Delta_0 q) + \mathbf{g}(\mathbf{a} \cdot \nabla_0 e) + e \mathbf{h}}. \quad (48)$$

The rest of the expressions are:

$$\nabla_0 m = \nabla_0 \left(\mu J^{-\frac{2}{3}} \right) = -\frac{2}{3} \mu J^{-\frac{5}{3}} \nabla_0 J = -\frac{2 \mu J^{-\frac{2}{3}}}{3 J} \mathbf{h} \Rightarrow \boxed{\nabla_0 m = -\frac{2 m}{3 J} \mathbf{h}}, \quad (49)$$

$$\begin{aligned} \nabla_0 I_1 &= \nabla_0 (3 + 2 \mathbf{a} \cdot \mathbf{g} + A G) = 2 \underbrace{(\nabla_0 \mathbf{a})^T \mathbf{g}}_{=0} + 2 \underbrace{(\nabla_0 \mathbf{g})^T \mathbf{a}}_{\mathbf{H}=\mathbf{H}^T \Rightarrow \mathbf{h}} + G \underbrace{\nabla_0 A}_{=0} + A \nabla_0 (\mathbf{g} \cdot \mathbf{g}), \\ &= 2 \mathbf{h} + 2 A \underbrace{(\nabla_0 \mathbf{g})^T \mathbf{g}}_{\mathbf{H}=\mathbf{H}^T \Rightarrow \mathbf{r}} \Rightarrow \boxed{\nabla_0 I_1 = 2 \mathbf{h} + 2 A \mathbf{r}}. \end{aligned} \quad (50)$$

$$\begin{aligned} \nabla_0 c &= \nabla_0 \left(\frac{\kappa}{2} (J^2 - 1) - m \frac{I_1}{3} \right) = \kappa J \nabla_0 J - \frac{I_1}{3} \nabla_0 m - \frac{m}{3} \nabla_0 I_1, \\ &\Rightarrow \boxed{\nabla_0 c = \kappa J \mathbf{h} - \frac{I_1}{3} \nabla_0 m - \frac{m}{3} \nabla_0 I_1}, \end{aligned} \quad (51)$$

$$\boxed{\nabla_0 d = \nabla_0 m + \nabla_0 c, \quad \nabla_0 e = \frac{1}{J} \nabla_0 c - \frac{c}{J^2} \mathbf{h}}. \quad (52)$$

5 Two-dimensional problem: plane-strain use of the 3D constitutive law

5.1 Important interpretation of the 2D case

The constitutive law in Eq. (20) is a three-dimensional law: it uses $J^{-2/3}$ and the three-dimensional deviatoric operator with the factor $1/3$. Therefore, to use *exactly the same material model* in a 2D problem, we adopt plane strain,

$$\boxed{u_z = 0, \quad \frac{\partial(\cdot)}{\partial z} = 0, \quad F_{zz} = 1}. \quad (53)$$

The tensors are still three-dimensional, although only the in-plane momentum equations are solved. In particular, σ_{zz} is generally nonzero. Replacing $J^{-2/3}$ by J^{-1} or replacing $1/3$ by $1/2$ would define a different, intrinsically two-dimensional constitutive model and would no longer reproduce Eq. (20).

5.2 Manufactured displacement and derivatives

Let

$$q(x, y, t) = \sin(\omega_x x) \sin(\omega_y y) \sin(\omega_t t), \quad (54)$$

with

$$\boxed{\mathbf{u}(x, y, t) = \begin{bmatrix} a_x q \\ a_y q \end{bmatrix}}. \quad (55)$$

For the three-dimensional plane-strain embedding we use

$$\mathbf{a} = \begin{bmatrix} a_x \\ a_y \\ 0 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} g_x \\ g_y \\ 0 \end{bmatrix}. \quad (56)$$

Introduce

$$s_x = \sin(\omega_x x), \quad c_x = \cos(\omega_x x), \quad s_y = \sin(\omega_y y), \quad c_y = \cos(\omega_y y), \quad s_t = \sin(\omega_t t), \quad c_t = \cos(\omega_t t). \quad (57)$$

Then

$$q = s_x s_y s_t, \quad g_x = \omega_x c_x s_y s_t, \quad g_y = \omega_y s_x c_y s_t. \quad (58)$$

The nonzero Hessian components are

$$H_{xx} = -\omega_x^2 q, \quad H_{yy} = -\omega_y^2 q, \quad H_{xy} = H_{yx} = \omega_x \omega_y c_x c_y s_t. \quad (59)$$

Moreover,

$$\Delta q = -(\omega_x^2 + \omega_y^2) q, \quad \ddot{q} = -\omega_t^2 q. \quad (60)$$

5.3 Deformation and strain tensors in 2D plane strain

The three-dimensional deformation gradient associated with the 2D field is

$$\mathbf{F} = \begin{bmatrix} 1 + a_x g_x & a_x g_y & 0 \\ a_y g_x & 1 + a_y g_y & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (61)$$

Its determinant is

$$J = 1 + a_x g_x + a_y g_y. \quad (62)$$

The inverse transpose is

$$\mathbf{F}^{-T} = \begin{bmatrix} \frac{1 + a_y g_y}{J} & -\frac{a_y g_x}{J} & 0 \\ -\frac{a_x g_y}{J} & \frac{1 + a_x g_x}{J} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (63)$$

Define

$$A = a_x^2 + a_y^2, \quad G = g_x^2 + g_y^2. \quad (64)$$

The right Cauchy–Green tensor is

$$\mathbf{C} = \begin{bmatrix} 1 + 2 a_x g_x + A g_x^2 & a_y g_x + a_x g_y + A g_x g_y & 0 \\ a_y g_x + a_x g_y + A g_x g_y & 1 + 2 a_y g_y + A g_y^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (65)$$

The Green–Lagrange strain is

$$\mathbf{E} = \frac{1}{2} \begin{bmatrix} 2 a_x g_x + A g_x^2 & a_y g_x + a_x g_y + A g_x g_y & 0 \\ a_y g_x + a_x g_y + A g_x g_y & 2 a_y g_y + A g_y^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (66)$$

The left Cauchy–Green tensor is

$$\mathbf{B} = \begin{bmatrix} 1 + 2 a_x g_x + G a_x^2 & a_x g_y + a_y g_x + G a_x a_y & 0 \\ a_x g_y + a_y g_x + G a_x a_y & 1 + 2 a_y g_y + G a_y^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (67)$$

The isochoric left Cauchy–Green tensor is simply

$$\bar{\mathbf{B}} = J^{-2/3} \mathbf{B}. \quad (68)$$

The first invariant is

$$I_1 = 3 + 2(a_x g_x + a_y g_y) + A G. \quad (69)$$

The inverse right Cauchy–Green tensor is

$$\mathbf{C}^{-1} = \begin{bmatrix} D_{xx} & D_{xy} & 0 \\ D_{xy} & D_{yy} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (70)$$

where

$$D_{xx} = 1 - \frac{2 a_x g_x}{J} + \frac{G a_x^2}{J^2}, \quad (71)$$

$$D_{yy} = 1 - \frac{2 a_y g_y}{J} + \frac{G a_y^2}{J^2}, \quad (72)$$

$$D_{xy} = -\frac{a_x g_y + a_y g_x}{J} + \frac{G a_x a_y}{J^2}. \quad (73)$$

5.4 Stress tensors in 2D

As before, with I_1 given by Eq. (69),

$$m = \mu J^{-2/3}, \quad c = \frac{\kappa}{2} (J^2 - 1) - \frac{m I_1}{3}, \quad d = m + c, \quad e = \frac{c}{J}. \quad (74)$$

The second Piola–Kirchhoff stress is

$$\mathbf{S} = \begin{bmatrix} m + c D_{xx} & c D_{xy} & 0 \\ c D_{xy} & m + c D_{yy} & 0 \\ 0 & 0 & d \end{bmatrix}. \quad (75)$$

The first Piola–Kirchhoff stress is

$$\mathbf{P} = \begin{bmatrix} d + (m - e) a_x g_x & m a_x g_y - e g_x a_y & 0 \\ m a_y g_x - e g_y a_x & d + (m - e) a_y g_y & 0 \\ 0 & 0 & d \end{bmatrix}. \quad (76)$$

The Kirchhoff stress is

$$\boldsymbol{\tau} = \begin{bmatrix} m B_{xx} + c & m B_{xy} & 0 \\ m B_{xy} & m B_{yy} + c & 0 \\ 0 & 0 & d \end{bmatrix}, \quad (77)$$

where B_{xx} , B_{xy} and B_{yy} are the in-plane components of $\mathbf{B}$ in Eq. (67). The Cauchy stress is

$$\boldsymbol{\sigma} = \frac{1}{J} \begin{bmatrix} m B_{xx} + c & m B_{xy} & 0 \\ m B_{xy} & m B_{yy} + c & 0 \\ 0 & 0 & d \end{bmatrix}. \quad (78)$$

The out-of-plane components $S_{zz} = P_{zz} = \tau_{zz} = d$ and $\sigma_{zz} = d/J$ are generally nonzero, but they do not depend on z and therefore do not enter the in-plane momentum equations.

5.5 Exact body force in 2D

The vectors $\mathbf{h} = \mathbf{H}\mathbf{a}$ and $\mathbf{r} = \mathbf{H}\mathbf{g}$ have in-plane components

$$h_x = H_{xx} a_x + H_{xy} a_y, \quad h_y = H_{xy} a_x + H_{yy} a_y, \quad (79)$$

$$r_x = H_{xx} g_x + H_{xy} g_y, \quad r_y = H_{xy} g_x + H_{yy} g_y. \quad (80)$$

Then, for $i = x, y$, it is possible to obtain $(\bullet)_{,i} = \frac{\partial(\bullet)}{\partial i}$ as

$$(I_1)_{,i} = 2 h_i + 2 A r_i, \quad (81)$$

$$m_{,i} = -\frac{2m}{3J} h_i, \quad (82)$$

$$c_{,i} = \kappa J h_i - \frac{I_1}{3} m_{,i} - \frac{m}{3} (I_1)_{,i}, \quad (83)$$

$$d_{,i} = m_{,i} + c_{,i}, \quad (84)$$

$$e_{,i} = \frac{c_{,i}}{J} - \frac{c}{J^2} h_i. \quad (85)$$

Define the two scalars

$$\boxed{R_{2D} = m \Delta q + g_x m_{,x} + g_y m_{,y}, \quad T_{2D} = a_x e_{,x} + a_y e_{,y}}. \quad (86)$$

The exact body force per unit reference volume is then $\mathbf{f}_0 = [f_{0x}, f_{0y}, f_{0z}]^T$:

$$\boxed{\begin{aligned} f_{0x} &= -\rho_0 \omega_t^2 a_x q - d_{,x} - a_x R_{2D} + g_x T_{2D} + e h_x, \\ f_{0y} &= -\rho_0 \omega_t^2 a_y q - d_{,y} - a_y R_{2D} + g_y T_{2D} + e h_y, \\ f_{0z} &= 0 \end{aligned}}. \quad (87)$$

5.6 Dirichlet boundary conditions in 2D

For

$$\Omega = [0, L_x] \times [0, L_y], \quad (88)$$

the exact displacement is prescribed on every side:

$$\mathbf{u}(0, y, t) = \mathbf{0}, \quad (89)$$

$$\mathbf{u}(L_x, y, t) = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \sin(\omega_x L_x) \sin(\omega_y y) \sin(\omega_t t), \quad (90)$$

$$\mathbf{u}(x, 0, t) = \mathbf{0}, \quad (91)$$

$$\mathbf{u}(x, L_y, t) = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \sin(\omega_x x) \sin(\omega_y L_y) \sin(\omega_t t). \quad (92)$$

If

$$\omega_x = \frac{n_x \pi}{L_x}, \quad \omega_y = \frac{n_y \pi}{L_y}, \quad n_x, n_y \in \mathbb{N}, \quad (93)$$

all boundary conditions are homogeneous Dirichlet conditions.

5.7 Initial conditions in 2D

Because $\sin(\omega_t t) = 0$ at $t = 0$,

$$\boxed{\mathbf{u}(x, y, 0) = \mathbf{0}}. \quad (94)$$

The initial velocity is nonzero:

$$\boxed{\dot{\mathbf{u}}(x, y, 0) = \omega_t \begin{bmatrix} a_x \\ a_y \end{bmatrix} \sin(\omega_x x) \sin(\omega_y y)}. \quad (95)$$

Also $\ddot{\mathbf{u}}(x, y, 0) = \mathbf{0}$. At the initial instant

$$\mathbf{F} = \mathbf{I}, \quad J = 1, \quad \mathbf{C} = \mathbf{B} = \mathbf{I}, \quad (96)$$

and the constitutive law is stress free:

$$\boxed{\mathbf{S} = \mathbf{P} = \boldsymbol{\tau} = \boldsymbol{\sigma} = \mathbf{0} \quad (t = 0)}. \quad (97)$$

6 Three-dimensional manufactured solution

6.1 Displacement field and derivatives

Let

$$q(x, y, z, t) = \sin(\omega_x x) \sin(\omega_y y) \sin(\omega_z z) \sin(\omega_t t), \quad (98)$$

and

$$\mathbf{u}(x, y, z, t) = \begin{bmatrix} a_x q \\ a_y q \\ a_z q \end{bmatrix}. \quad (99)$$

Define

$$s_z = \sin(\omega_z z), \quad c_z = \cos(\omega_z z), \quad (100)$$

with $s_x, c_x, s_y, c_y, s_t, c_t$ defined as before. Then

$$\mathbf{g} = \begin{bmatrix} \omega_x c_x s_y s_z s_t \\ \omega_y s_x c_y s_z s_t \\ \omega_z s_x s_y c_z s_t \end{bmatrix}. \quad (101)$$

The Hessian is

$$\mathbf{H} = \begin{bmatrix} -\omega_x^2 q & H_{xy} & H_{xz} \\ H_{xy} & -\omega_y^2 q & H_{yz} \\ H_{xz} & H_{yz} & -\omega_z^2 q \end{bmatrix}, \quad (102)$$

where

$$H_{xy} = \omega_x \omega_y c_x c_y s_z s_t, \quad (103)$$

$$H_{xz} = \omega_x \omega_z c_x s_y c_z s_t, \quad (104)$$

$$H_{yz} = \omega_y \omega_z s_x c_y c_z s_t. \quad (105)$$

Moreover,

$$\Delta q = -(\omega_x^2 + \omega_y^2 + \omega_z^2) q, \quad \ddot{q} = -\omega_t^2 q. \quad (106)$$

6.2 Deformation and strain tensors in 3D

The deformation gradient is

$$\mathbf{F} = \begin{bmatrix} 1 + a_x g_x & a_x g_y & a_x g_z \\ a_y g_x & 1 + a_y g_y & a_y g_z \\ a_z g_x & a_z g_y & 1 + a_z g_z \end{bmatrix}. \quad (107)$$

Its Jacobian is

$$J = 1 + a_x g_x + a_y g_y + a_z g_z. \quad (108)$$

The inverse transpose is

$$\mathbf{F}^{-T} = \mathbf{I} - \frac{1}{J} \begin{bmatrix} g_x a_x & g_x a_y & g_x a_z \\ g_y a_x & g_y a_y & g_y a_z \\ g_z a_x & g_z a_y & g_z a_z \end{bmatrix}. \quad (109)$$

Define

$$A = a_x^2 + a_y^2 + a_z^2, \quad G = g_x^2 + g_y^2 + g_z^2. \quad (110)$$

The right Cauchy–Green tensor is

$$\mathbf{C} = \begin{bmatrix} C_{xx} & C_{xy} & C_{xz} \\ C_{xy} & C_{yy} & C_{yz} \\ C_{xz} & C_{yz} & C_{zz} \end{bmatrix}, \quad (111)$$

where

$$C_{xx} = 1 + 2 a_x g_x + A g_x^2, \quad C_{xy} = a_y g_x + a_x g_y + A g_x g_y, \quad (112)$$

$$C_{yy} = 1 + 2 a_y g_y + A g_y^2, \quad C_{xz} = a_z g_x + a_x g_z + A g_x g_z, \quad (113)$$

$$C_{zz} = 1 + 2 a_z g_z + A g_z^2, \quad C_{yz} = a_z g_y + a_y g_z + A g_y g_z. \quad (114)$$

The Green–Lagrange strain is

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}). \quad (115)$$

The left Cauchy–Green tensor is

$$\mathbf{B} = \begin{bmatrix} B_{xx} & B_{xy} & B_{xz} \\ B_{xy} & B_{yy} & B_{yz} \\ B_{xz} & B_{yz} & B_{zz} \end{bmatrix}, \quad (116)$$

where

$$B_{xx} = 1 + 2 a_x g_x + G a_x^2, \quad B_{xy} = a_x g_y + a_y g_x + G a_x a_y, \quad (117)$$

$$B_{yy} = 1 + 2 a_y g_y + G a_y^2, \quad B_{xz} = a_x g_z + a_z g_x + G a_x a_z, \quad (118)$$

$$B_{zz} = 1 + 2 a_z g_z + G a_z^2, \quad B_{yz} = a_y g_z + a_z g_y + G a_y a_z. \quad (119)$$

The isochoric tensor is

$$\overline{\mathbf{B}} = J^{-2/3} \mathbf{B}, \quad (120)$$

and the first invariant is

$$I_1 = 3 + 2(a_x g_x + a_y g_y + a_z g_z) + A G. \quad (121)$$

The inverse right Cauchy–Green tensor is

$$\mathbf{C}^{-1} = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{xy} & D_{yy} & D_{yz} \\ D_{xz} & D_{yz} & D_{zz} \end{bmatrix}, \quad (122)$$

with

$$D_{xx} = 1 - \frac{2 a_x g_x}{J} + \frac{G a_x^2}{J^2}, \quad D_{xy} = -\frac{a_x g_y + a_y g_x}{J} + \frac{G a_x a_y}{J^2}, \quad (123)$$

$$D_{yy} = 1 - \frac{2 a_y g_y}{J} + \frac{G a_y^2}{J^2}, \quad D_{xz} = -\frac{a_x g_z + a_z g_x}{J} + \frac{G a_x a_z}{J^2}, \quad (124)$$

$$D_{zz} = 1 - \frac{2 a_z g_z}{J} + \frac{G a_z^2}{J^2}, \quad D_{yz} = -\frac{a_y g_z + a_z g_y}{J} + \frac{G a_y a_z}{J^2}. \quad (125)$$

6.3 Stress tensors in 3D

As before,

$$m = \mu J^{-2/3}, \quad c = \frac{\kappa}{2}(J^2 - 1) - \frac{m I_1}{3}, \quad d = m + c, \quad e = \frac{c}{J}. \quad (126)$$

The second Piola–Kirchhoff stress is

$$\mathbf{S} = \begin{bmatrix} m + c D_{xx} & c D_{xy} & c D_{xz} \\ c D_{xy} & m + c D_{yy} & c D_{yz} \\ c D_{xz} & c D_{yz} & m + c D_{zz} \end{bmatrix}. \quad (127)$$

The first Piola–Kirchhoff stress is

$$\mathbf{P} = \begin{bmatrix} d + (m - e) a_x g_x & m a_x g_y - e g_x a_y & m a_x g_z - e g_x a_z \\ m a_y g_x - e g_y a_x & d + (m - e) a_y g_y & m a_y g_z - e g_y a_z \\ m a_z g_x - e g_z a_x & m a_z g_y - e g_z a_y & d + (m - e) a_z g_z \end{bmatrix}. \quad (128)$$

The Kirchhoff stress is

$$\boldsymbol{\tau} = \begin{bmatrix} m B_{xx} + c & m B_{xy} & m B_{xz} \\ m B_{xy} & m B_{yy} + c & m B_{yz} \\ m B_{xz} & m B_{yz} & m B_{zz} + c \end{bmatrix}. \quad (129)$$

The Cauchy stress is

$$\boldsymbol{\sigma} = \frac{1}{J} \begin{bmatrix} m B_{xx} + c & m B_{xy} & m B_{xz} \\ m B_{xy} & m B_{yy} + c & m B_{yz} \\ m B_{xz} & m B_{yz} & m B_{zz} + c \end{bmatrix}. \quad (130)$$

6.4 Exact body force in 3D

The vector $\mathbf{h} = \mathbf{H}\mathbf{a}$ has components

$$h_x = -\omega_x^2 q a_x + H_{xy} a_y + H_{xz} a_z, \quad (131)$$

$$h_y = H_{xy} a_x - \omega_y^2 q a_y + H_{yz} a_z, \quad (132)$$

$$h_z = H_{xz} a_x + H_{yz} a_y - \omega_z^2 q a_z. \quad (133)$$

The vector $\mathbf{r} = \mathbf{H}\mathbf{g}$ has components

$$r_x = -\omega_x^2 q g_x + H_{xy} g_y + H_{xz} g_z, \quad (134)$$

$$r_y = H_{xy} g_x - \omega_y^2 q g_y + H_{yz} g_z, \quad (135)$$

$$r_z = H_{xz} g_x + H_{yz} g_y - \omega_z^2 q g_z. \quad (136)$$

For $i = x, y, z$, it is possible to obtain $(\bullet)_{,i} = \frac{\partial(\bullet)}{\partial i}$ as

$$(I_1)_{,i} = 2 h_i + 2 A r_i, \quad (137)$$

$$m_{,i} = -\frac{2m}{3J} h_i, \quad (138)$$

$$c_{,i} = \kappa J h_i - \frac{I_1}{3} m_{,i} - \frac{m}{3} (I_1)_{,i}, \quad (139)$$

$$d_{,i} = m_{,i} + c_{,i}, \quad (140)$$

$$e_{,i} = \frac{c_{,i}}{J} - \frac{c}{J^2} h_i. \quad (141)$$

Define

$$R_{3D} = m \Delta q + g_x m_{,x} + g_y m_{,y} + g_z m_{,z}, \quad (142)$$

and

$$T_{3D} = a_x e_{,x} + a_y e_{,y} + a_z e_{,z}. \quad (143)$$

Then the exact body-force vector is

$$\begin{bmatrix} f_{0x} = -\rho_0 \omega_t^2 a_x q - d_{,x} - a_x R_{3D} + g_x T_{3D} + e h_x, \\ f_{0y} = -\rho_0 \omega_t^2 a_y q - d_{,y} - a_y R_{3D} + g_y T_{3D} + e h_y, \\ f_{0z} = -\rho_0 \omega_t^2 a_z q - d_{,z} - a_z R_{3D} + g_z T_{3D} + e h_z \end{bmatrix}. \quad (144)$$

6.5 Dirichlet boundary conditions in 3D

For

$$\Omega = [0, L_x] \times [0, L_y] \times [0, L_z], \quad (145)$$

$\mathbf{u} = \mathbf{u}^{\text{ex}}$ is imposed on all six faces:

$$\mathbf{u}(0, y, z, t) = \mathbf{0}, \quad (146)$$

$$\mathbf{u}(L_x, y, z, t) = \mathbf{a} \sin(\omega_x L_x) \sin(\omega_y y) \sin(\omega_z z) \sin(\omega_t t), \quad (147)$$

$$\mathbf{u}(x, 0, z, t) = \mathbf{0}, \quad (148)$$

$$\mathbf{u}(x, L_y, z, t) = \mathbf{a} \sin(\omega_x x) \sin(\omega_y L_y) \sin(\omega_z z) \sin(\omega_t t), \quad (149)$$

$$\mathbf{u}(x, y, 0, t) = \mathbf{0}, \quad (150)$$

$$\mathbf{u}(x, y, L_z, t) = \mathbf{a} \sin(\omega_x x) \sin(\omega_y y) \sin(\omega_z L_z) \sin(\omega_t t). \quad (151)$$

If

$$\omega_i = \frac{n_i \pi}{L_i}, \quad n_i \in \mathbb{N}, \quad i = x, y, z, \quad (152)$$

all six conditions are homogeneous Dirichlet conditions.

6.6 Initial conditions in 3D

The exact initial displacement is

$$\boxed{\mathbf{u}(x, y, z, 0) = \mathbf{0}}. \quad (153)$$

The initial velocity is

$$\boxed{\dot{\mathbf{u}}(x, y, z, 0) = \omega_t \mathbf{a} \sin(\omega_x x) \sin(\omega_y y) \sin(\omega_z z)}. \quad (154)$$

Also $\ddot{\mathbf{u}}(x, y, z, 0) = \mathbf{0}$. Since $\mathbf{g} = \mathbf{0}$ at $t = 0$,

$$\mathbf{F} = \mathbf{I}, \quad J = 1, \quad \mathbf{C} = \mathbf{B} = \mathbf{I}, \quad (155)$$

and therefore

$$\boxed{\mathbf{S} = \mathbf{P} = \boldsymbol{\tau} = \boldsymbol{\sigma} = \mathbf{0} \quad (t = 0)}. \quad (156)$$

7 Implementation summary

For each point $\mathbf{X}$ and time t , a direct implementation of the MMS can follow this sequence:

1. Evaluate q , $\mathbf{g} = \nabla_0 q$, $\mathbf{H} = \nabla_0 \mathbf{g}$, $\Delta_0 q$, and $\ddot{q}$: Eqs. (55)–(60) in 2D and Eqs. (99)–(106) in 3D. In 2D, use the plane-strain embedding $a_z = g_z = 0$ and keep all tensors 3×3 .
2. Evaluate $\mathbf{F} = \mathbf{I} + \mathbf{a} \otimes \mathbf{g}$ and $J = 1 + \mathbf{a} \cdot \mathbf{g}$ from Eq. (28) and Eqs. (62) or (108). Check that $J > 0$.
3. Evaluate $A = \mathbf{a} \cdot \mathbf{a}$, $G = \mathbf{g} \cdot \mathbf{g}$, and $I_1 = 3 + 2 \mathbf{a} \cdot \mathbf{g} + A G$ from Eq. (39). The constant is 3 also in 2D (Eq. (69)), because $C_{zz} = 1$.
4. Evaluate the scalars of Eqs. (23) and (40),

$$m = \mu J^{-2/3}, \quad c = \frac{\kappa}{2}(J^2 - 1) - \frac{m I_1}{3}, \quad d = m + c, \quad e = \frac{c}{J}.$$

5. Construct the required stress measure with Eq. (24),

$$\mathbf{S} = m \mathbf{I} + c \mathbf{C}^{-1}, \quad \mathbf{P} = m \mathbf{F} + c \mathbf{F}^{-T}, \quad \boldsymbol{\tau} = m \mathbf{B} + c \mathbf{I}, \quad \boldsymbol{\sigma} = J^{-1} \boldsymbol{\tau},$$

where $\mathbf{F}^{-T}$, $\mathbf{B}$ and $\mathbf{C}^{-1}$ are given by Eqs. (32), (36) and (38). Equivalently, $\mathbf{P} = d \mathbf{I} + m \mathbf{a} \otimes \mathbf{g} - e \mathbf{g} \otimes \mathbf{a}$ (Eq. (41)). In components, see Eqs. (75)–(78) in 2D and Eqs. (127)–(130) in 3D.

6. Evaluate $\mathbf{h} = \mathbf{H} \mathbf{a} = \nabla_0 J$, $\mathbf{r} = \mathbf{H} \mathbf{g}$, and the scalar gradients $\nabla_0 m$, $\nabla_0 I_1$, $\nabla_0 c$, $\nabla_0 d$ and $\nabla_0 e$ from Eqs. (49)–(52).
7. Evaluate $\nabla_0 \cdot \mathbf{P}$ from Eq. (47) and then $\mathbf{f}_0$ from Eq. (48). In components, see Eq. (87) in 2D (where $f_{0z} = 0$) and Eq. (144) in 3D.

8. Impose $\mathbf{u} = \mathbf{u}^{\text{ex}}$ on the complete boundary, Eq. (4), and use the exact initial displacement and velocity, Eqs. (94)–(95) in 2D and Eqs. (153)–(154) in 3D.

The body force in this report is per unit *reference* volume. If a solver instead uses the spatial equation

$$\rho \ddot{\mathbf{u}} = \nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma} + \mathbf{f}, \quad (157)$$

then $\rho = \rho_0/J$ and the corresponding body force per unit current volume is

$$\boxed{\mathbf{f} = \frac{1}{J} \mathbf{f}_0}. \quad (158)$$